

Lakshay Chauhan

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EDUCATION

Indraprastha Institute of Information Technology (IIIT) Delhi Jul 2021 – Jun 2025
Bachelor of Technology in Computer Science and Engineering Delhi, India

WORK EXPERIENCE

Member of Technical Staff Jul 2025 – Present
ZL Technologies, Inc / [Website](#) Hyderabad, India

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

Research Assistant | [Gitlab Project Organization](#) May 2024 – May 2025
Networks and Systems Security Lab - IIITD / [Website](#) Delhi, India

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

PROJECTS

Quorum — Distributed Key-Value Store | [Repository](#) Jan 2026 – Mar 2026

- Built a distributed key-value store in Java supporting concurrent client requests and inter-node communication for scalable data storage.
- Designed core distributed system components including request routing, replication mechanisms, and fault-tolerant data handling. Applied concepts of networking, synchronization, and scalable backend architecture.

AI Customer Service Agent | [Repository](#) May 2025 – Jul 2025

- Developed an end-to-end AI-powered customer service agent with real-time voice and chat capabilities using LiveKit and Python.
- Implemented continuously evolving knowledge-base integration and asynchronous workflows for contextual response generation. Worked with event-driven systems, LLM integration, and scalable backend orchestration.

JDBC Transaction-Based Locking Mechanism for Java | [Repository](#) Feb 2026 – Mar 2026

- Implemented a database-backed distributed locking mechanism in Java using JDBC transactions and row-level locks.
- Enabled synchronization across threads, processes, and multiple application instances while ensuring transactional consistency and concurrency safety. Explored concepts such as isolation levels, deadlock prevention, and reliable coordination in distributed environments.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Rust, JavaScript, SQL, Bash

Frameworks & Technologies: ReactJS, Django, PyTorch, REST APIs, gRPC, LiveKit, Raylib

Databases: PostgreSQL, SQLite3, Elasticsearch, MySQL, MongoDB

Developer Tools & Platforms: Git, Linux, Docker, Emacs, Gradle, Maven, Google Cloud Platform (GCP), GDB, OpenLiteSpeed

Core Concepts: Data Structures & Algorithms, Distributed Systems, Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming, Concurrency & Synchronization, Compiler Design, Machine Learning

AWARDS

- Awarded the **Summer Undergraduate Research Fellowship** (2023) by **IRD-IIITD** for the project “Utilizing Ultrasonic Distance Sensors as a Mapping Tool to Design User-Friendly CST”.
- Awarded the **CHANAKYA Fellowship** (2024) by **iHub Anubhuti Foundation** for the project “A Unified Approach to User Emotion Detection through Emojis and Textual Analysis”.